Lean formalization of an Assumptions of Physics technical brief: log

This log records which statements of the technical brief in this directory were checked in Lean, where the formal version differs from the brief, and any problems found in the brief. The brief is available at `https://assumptionsofphysics.org/resources/briefs`.

Setup

- Lean `v4.36.0-rc1`; Mathlib at commit `2cabef97`; physlib (`master`) and TauCeti (commit `eb4dd24`, on the same Mathlib) as further dependencies.
- Build: `lake build` in the project root.
- Every file is checked with `#print axioms` on its main results. A result is recorded here only if it depends on no axioms beyond `propext`, `Classical.choice` and `Quot.sound`, and so has no `sorry`.
- Imports are kept to the Mathlib modules actually needed.
- Material that belongs in Mathlib is kept in `ForMathlib/`, in Mathlib's directory layout (library `ForMathlib`), and imported by the brief files.

1 Superpositions correspond to multiple decompositions

Files.

- File: `briefs/SuperpositionAndMultipleMixtures/SuperpositionAndMultipleMixtures.lean`
- Library: `SuperpositionAndMultipleMixtures`
- Imports: `ForPhyslib.QuantumMechanics.Wigner.Density` (for `proj`, $|\psi\rangle\langle\psi|$, and its parallelogram law), `ForMathlib.Analysis.InnerProductSpace.Projection.FiniteDimensional`
- New `ForMathlib` material: `ForMathlib/Analysis/InnerProductSpace/Projection/FiniteDimensional.lean` (`Submodule.mem_span_of_forall_inner_eq_zero`: a vector orthogonal to everything that is orthogonal to a finite set lies in its span).

Setting. E is a complex inner product space of any dimension, not necessarily complete. For a vector ψ the pure state is `pureState` $\psi = \rho_\psi = |\psi\rangle\langle\psi| \,/\, \langle\psi|\psi\rangle$ (it is 0 at $\psi = 0$), and `mixture` $p\ \psi\ q\ \chi = p\rho_\psi + q\rho_\chi$. An operator is mixed (`IsMixed`) if it is not ρ_ψ for any $\psi \neq 0$. The conditions are:

- `QSupPur` $\phi\ \psi_1\ \psi_2$: ψ_1, ψ_2 linearly independent and $\phi = c_1\psi_1 + c_2\psi_2$ with $c_1, c_2 \neq 0$;
- `QSupMix` $\phi\ \psi_1\ \psi_2$: there are $p_1, p_2, p_\phi, p_{\hat\phi} \in (0,1)$ with $p_1 + p_2 = p_\phi + p_{\hat\phi} = 1$ and a state $\hat\phi \neq 0$ such that $\rho = p_1\rho_{\psi_1} + p_2\rho_{\psi_2}$ is mixed, $\rho = p_\phi\rho_\phi + p_{\hat\phi}\rho_{\hat\phi}$, and $\rho_\phi \notin \{\rho_{\psi_1}, \rho_{\psi_2}\}$.

Results.

Brief	Lean
eq. (2), $\|\phi\rangle\langle\phi\| + \|\hat\phi\rangle\langle\hat\phi\| = 2\|c_1\|^2\|\psi_1\rangle\langle\psi_1\| + 2\|c_2\|^2\|\psi_2\rangle\langle\psi_2\|$	`proj_add_proj_hat`
eqs. (3), (4), QSUP-PUR ⇒ QSUP-MIX	`QSupPur.qSupMix`, with the weights of eq. (3)
"it is supported by a two dimensional subspace"	`mem_span_of_mixture_eq`: if $q\rho_\phi + q'\rho_{\phi'} = p_1\rho_{\psi_1} + p_2\rho_{\psi_2}$ with $q > 0$, $q' \geq 0$, then $\phi \in \mathrm{span}\{\psi_1, \psi_2\}$
QSUP-MIX ⇒ QSUP-PUR	`QSupMix.qSupPur`, for $\phi \neq 0$
Theorem 1	`qSupPur_iff_qSupMix`, for $\phi \neq 0$
(used for QSUP-MIX)	`isMixed_mixture`: a mixture of two linearly independent states with positive weights is not pure

Differences from the brief.

- The equivalence is proved for each triple ϕ, ψ_1, ψ_2 with $\phi \neq 0$; no other vector needs to be nonzero, and E need not be complete or finite-dimensional.
- The brief writes ρ as a density operator. The formalization does not use the trace: ρ is a convex combination of two pure states, which is all the argument needs.
- QSUP-MIX ⇒ QSUP-PUR uses neither that ρ is mixed nor that $p_{\hat\phi} > 0$; only $p_\phi > 0$, $p_{\hat\phi} \geq 0$ and $\rho_\phi \notin \{\rho_{\psi_1}, \rho_{\psi_2}\}$.

Problems found in the brief. None that affect the theorem; the demonstration leaves out some steps, which the formalization supplies.

- QSUP-PUR ⇒ QSUP-MIX: the brief does not check that $\hat\phi \neq 0$ (needed to form $\rho_{\hat\phi}$), that $\rho_\phi \notin \{\rho_{\psi_1}, \rho_{\psi_2}\}$, or that ρ is mixed. All three follow from the linear independence of ψ_1, ψ_2 and $c_1, c_2 \neq 0$.
- QSUP-MIX ⇒ QSUP-PUR: "Since ρ is the mixture of two pure states, it is supported by a two dimensional subspace" presupposes that ψ_1 and ψ_2 are linearly independent, which QSUP-MIX does not state, and the brief does not show that $c_1, c_2 \neq 0$. Both follow from $\rho_\phi \notin \{\rho_{\psi_1}, \rho_{\psi_2}\}$: if ψ_2 were a multiple of ψ_1, or c_1 or c_2 were 0, then ϕ would be a multiple of ψ_1 or of ψ_2. The support claim itself is proved from the quadratic form: a vector orthogonal to ψ_1 and ψ_2 gives $\langle x|\rho|x\rangle = 0$, hence $\langle\phi|x\rangle = 0$.